

Shin Khant Soe Moe

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Computer Science graduate from Newcastle University with a predicted first-class degree, with strong experience in full-stack development, cloud-native applications, and HCI research. Built and deployed production-ready web applications end-to-end, from database design to CI/CD pipelines.

Education

BSc (Hons) Computer Science, Newcastle University 2023 – 2026

- Achieved 1st class in Year 1 (85.5%) and Year 2 (75.5%); on track to graduate with 1st class honours.
- Relevant modules: Gaming Technologies and Simulations, Data Visualisation and Visual Analytics, Distributed Systems, Security Programming, Algorithm Design and Analysis, Software Systems Design and Implementation.

Technical skills

Programming Languages: C#, Python, Java, JavaScript, React.js, [D3.js](#)

Frameworks and Tools: Git, Github, Docker, Azure, Python Flask, VS Code, Supabase, PostgreSQL

Game Engines: Unity, Godot, Unreal

Softwares and Apps: Microsoft Office, Adobe Premiere Pro, Inkscape, Aseprite, Canva

Projects

Final Year Dissertation: Cross-Device Data Dashboard & User Study 2025 – 2026

- Designed and developed a cross-device data dashboard in React and D3.js, deployed across a large touch display and a mobile phone, to study how display size shapes exploratory data analysis.
- Conducted a within-subjects HCI study with 11 participants, capturing 4,844 instrumented interaction events alongside SUS, NASA-TLX, and qualitative survey data to triangulate usability, cognitive load, and user experience.
- Performed mixed-methods analysis using paired Wilcoxon signed-rank tests and Braun & Clarke's reflexive thematic analysis to characterise design trade-offs across display sizes.
- Found a notable divergence between subjective preference and objective performance: the touch display received significantly higher SUS scores and stronger qualitative endorsement despite no significant difference in task correctness or completion time.

Chalklist — Full-Stack Task Management App with DevOps Pipeline

- Built a multi-user to-do web application using Python Flask and Supabase, following a three-tier architecture with clear separation between presentation, application, and data layers.
- Implemented database-level access control using Supabase Row Level Security, enforcing user data isolation at the PostgreSQL layer rather than relying solely on application-level filtering.
- Adopted a GitFlow branching strategy with GitHub Actions for CI, and deployed to Microsoft Azure with structured logging and centralised observability tooling.

Turn-Based Combat System

- Built a 4v4 turn-based RPG combat system in Godot 4, implementing a state-machine-driven battle loop handling player and enemy phases.
- Designed a resource-based architecture where skills, status effects, passives, and weapons are fully decoupled data objects, allowing new characters and abilities to be configured without modifying core logic.
- Implemented a stackable status effect system supporting decaying, permanent, and permanent-stackable types, alongside a per-character energy economy and ultimate ability buildup mechanic.
- Decoupled game logic from UI entirely using a signal-based event bus, keeping the battle manager stateless with respect to rendering.

3D Third-Person Game Tech Demo

- Developed a third-person game tech demo in Unity to explore advanced game engineering — physics-driven projectiles, dynamic physics materials, and gravity-altering and time-scaling power-ups.
- Implemented multiple enemy types using NavMeshAgents and finite state machines with probabilistic transitions, planning-based decision making, and predictive dodge evaluation using simulated outcomes.
- Designed a central game manager and token-based enemy spawner to dynamically scale difficulty based on player state.
- Explored environment systems including dynamic NavMesh obstacles, procedural geometry changes, and infinite tiled environments.

Personal Portfolio Website

- Designed and developed a personal portfolio website from scratch using React and Vite, featuring a custom CSS design system and fully responsive layout across desktop and mobile.
- Integrated Formspree for functional contact form submission and implemented smooth section navigation, CV download, and social links.

Additional Experiences

Co-founder & Education Coordinator — Newcastle GameDev Society 2024 – 2026

- Co-founded the society in second year alongside the president, building it into a welcoming community of ~50 active members annually that continues independently beyond my graduation.
- Hosted 4–5 workshops and 4–5 game jams per year, drawing 15–20 attendees per workshop and 5+ jam entries, teaching game development fundamentals to members at all experience levels.
- Served on the committee as Education Coordinator for two years, designing and delivering the society's educational programme.

Game Jam — Newcastle GameDev Society 2024

- Designed and developed a 2D speedrunning game in Godot within a 72-hour jam alongside a collaborator, mentoring them through their first game development project while shipping a fully playable demo on itch.io.

Video Editor (Volunteer) — National Blood Center Yangon 2023

- Produced a 10-minute educational video on blood types and blood donation as part of a volunteer team, handling editing, motion graphics, and post-production.